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JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science & Engineering

"A Place to Learn, A Chance to Grow"

Session: 2025-26

Education to Eternity

MSE-1 EXAMINATION

Course: B. Tech. in Computer Science & Engineering **Semester/ Year:** III Semester/ 2nd Year

Subject: Analytical Skills and Ethical Application **Subject Code:** CS3V001

Max Marks: 30 **Date: 06/08/2025** **Duration:- 1 Hr.**

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

1. Illustrate the use of number system rules to solve basic simplification problems.
2. To analyze and solve real-life and mathematical problems involving percentages, including population variations and percentage-based changes.
3. To apply and evaluate profit and loss concepts to solve problems involving cost price, selling price, marked price, and discounts effectively.
4. To apply formulas for simple and compound interest to solve practical and mixed-interest problems.
5. To analyze mathematical concepts related to ratios, proportions, partnerships, and age problems to solve both theoretical and practical scenarios.

		Level/CO	Marks
	Objective type questions (Compulsory Question)		
Q.1	If 30% of a number is added to 40, it matches precisely 50% of the same number. What is this number that balances the equation? a.60 b. 80 c. 100 d. 200	L3/CO2	1
Q.2	Two consecutive even numbers, when squared and added together, give a total of 244. What are these two numbers? a.10, 12 b. 12, 14 c. 14, 16 d. 16, 20	L2/CO1	1
Q.3	What is the bigger Number by which dividing 1057, 1259, 1661 gives the remainder of 7, 9 and 11 respectively? a. 50 b. 51 c. 49 d. 48	L3/CO1	1
Q.4	What is HCF of 36/75 , 48/150 , 72/135? a. 12/144 b. 144/5 c. 12/1350 d. 1350/12	L2/CO1	1
Q.5	What is LCM of 36/225, 48/150, 72/65? a.5/144 b. 144/5 c. 72/5 d. 15/72	L2/CO1	1
Q.6	Three bells ring at intervals of 36 sec, 40 sec and 48 sec respectively. They start bringing together at a particular time. It will ring together after every? a. 6min b. 12min c. 18min d. 24min	L3/CO1	1
Q.7	$8\frac{1}{6} + 5\frac{1}{8} + 4\frac{2}{3} = ?$ a. $17\frac{7}{8}$ b. $16\frac{1}{7}$ c. $15\frac{2}{3}$ d. $17\frac{1}{7}$	L3/CO1	1
Q.8	831 - 31.425 + 18.749 -? = 12 a. 3.155 b. 3.105 c. 4.155 d. 4.345	L2/CO1	1

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Q.9	Find LCM of 1/3, 5/6, 5/9, 10/27 a. 10/3 b. 5/3 c. 5/27 d. 10/27	L2/CO1	1
Q.10	Find out the largest number from which divide 134, 59, 184 and leaves remainder 9 in each case? a. 20 b. 23 c. 24 d. 25	L3/CO1	1
Q.11	$3^{11}+3^{12}+3^{13}+3^{14}$ is divisible by a. 7 b. 8 c. 11 d. 14	L4/CO1	1
Q.12	$5\frac{1}{5}+4\frac{1}{2}+4\frac{1}{3}=?$ a. 14(1/10) b. 13(1/5) c. 13/30 d. 14(1/30)	L2/CO1	1
Q.13	If 60% of a number is 90, what is 80% of the number? a. 100 b. 120 c. 110 d. 150	L3/CO2	1
Q.14	A's salary is 25% more than B's. If B's salary is ₹24,000, what is A's salary? a. ₹30,000 b. ₹29,000 c. ₹28,000 d. ₹31,000	L4/CO2	1
Q.15	A machine's value depreciates by 10% annually. What will be its value after 2 years if the present value is ₹50,000? a. ₹40500 b. ₹45000 c. ₹42000 d. ₹48000	L4/CO2	1
Q.16	The population of a town increased by 14⅔% in the 1st year and then by 25% in the 2nd year. If the population after 2 years becomes 4100, find the original population. a. 2800 b. 2900 c. 3000 d. 3200	L3/CO2	1
Q.17	In an election between two candidates, the winning candidate got 70% of the votes polled and won by 15,400 votes. What is the number of votes polled for the losing candidate? a. 66,000 b. 55,000 c. 66,600 d. 60,000	L4/CO2	1
Q.18	If the price of tea is increased by 20%, by how much per cent the consumption of tea be reduced so that there is no increase in the expenditure on it? a. 83 1/3% b. 20% c.16 2/3% d.8 1/3%	L3/CO2	1
Q.19	Evaluate the expression: $(48 \div 12 \times 2 + (3 \times 7 + 9)) = ?$ a.36 b. 30 c. 38 d. 42	L2/CO1	1
Q.20	If: $(2^4) \times (4^4) \times (16^2) \times 32 = 2^x$ Then what is the value of x? a.36 b. 44 c. 40 d. 48	L3/CO1	1
Q.21	$\sqrt[3]{2197} \times \sqrt[3]{729} = ?$ a. 117 b. 91 c. 126 d. 99	L2/CO1	1
Q.22	$\sqrt{1296} \times \sqrt{1764} + ? = (39)$ ^ 2 a.9 b. 7 c. 22 d. 18	L2/CO1	1
Q.23	Find the greatest number which divide 403,527, 589 leaving no remainder? a. 27 b. 29 c. 30 d. 31	L3/CO1	1
Q.24	Imagine a two-digit number where the sum of its digits equals 9. Furthermore, the difference between these digits is 3. What is the number?	L2/CO1	1

	a.63 b. 36 c. 54 d. 45		
Q.25	Think of a number such that when 15 is subtracted from seven times the number, the result equals 10 more than twice the number. What could this magical number be?	L3/CO1	1
	a. 5 b. 6 c. 7 d. 8		
Q.26	Two numbers have a product of 120, and their greatest common divisor (HCF) is 6. Find their least common multiple (LCM)?	L3/CO1	1
	a. 20 b. 30 c. 40 e. 50		
Q.27	Consider a two-digit number where the sum of its digits equals 12, and their difference is 4. Find the number?	L2/CO1	1
	a.48 b. 39 c. 84 d. 57		
Q.28	If a - b = 3 and a^2 + b^2 = 29, find the value of ab.	L2/CO1	1
	a.10 b. 12 c. 15 d. 18		
Q.29	If 432P1 is completely divisible by '9', then what is the value of 'P'?	L4/CO1	1
	a.1 b. 8 c.7 d. 2		
Q.30	If 30% of a number is added to 40, it matches precisely 50% of the same number. What is this number that balances the equation?	L4/CO2	1
	a. 60 b. 80 c. 100 d.200		